Fresh concrete: workability and its tests

Notes from the lecture slides “Module 6 – Fresh Concrete”

In this module

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1 Workability

1.1 Definition

Workability is the **ease and homogeneity** with which freshly mixed concrete or mortar can be **mixed, placed, compacted and finished**.

- The word **consistency** is sometimes used for the same idea.
- Workability is a measure of the **fluidity or mobility** of fresh concrete.
- There is no single "correct" workability. The workability needed depends on the **job conditions, weather, type of work and method of compaction**. For example, a heavily reinforced thin wall needs a much more workable mix than a road slab compacted by a power vibrator.

Why workability matters

Concrete that is too stiff for the job cannot be fully compacted, so **voids remain**. **Voids reduce the density, and lower density means lower strength.** Workability is therefore not just about convenience: it decides whether the designed strength is actually achieved.

1.2 Strength ratio vs density ratio

The slide's graph plots

- **density ratio** = density actually achieved ÷ density when fully compacted, against
- **strength ratio** = strength actually achieved ÷ strength when fully compacted.

Density ratio	Approximate strength ratio
1.00 (fully compacted)	1.00
0.95 (5% voids)	about 0.7
0.90 (10% voids)	about 0.45–0.5
0.85	about 0.3
0.80	about 0.2
0.75	about 0.1

The curve is steep: **just 5% voids can cut strength by about 30%**. Full compaction is essential, and it needs adequate workability.

2 Factors affecting workability

Factor	Effect on workability	Why
Water content (kg/m^3)	**Most important; more water → more workable** (direct)	Water lubricates the particles. But extra water also raises the w/c ratio and lowers strength

Factor	Effect on workability	Why
Aggregate type and grading	Fine, **irregular, angular, rough or flaky** particles → **less workable** (more water needed)	More surface area to coat; more interlock and friction. Rounded, smooth, well-graded aggregate gives better workability
Aggregate/cement ratio	Higher a/c (leaner mix) → **less workable** (inverse)	Less paste is available to coat and lubricate the aggregate
Admixtures	Plasticizers and superplasticizers → **more workable**	Disperse the cement grains (Module 3)
Fineness of cement	Finer cement → slightly **less workable** for the same water	More surface area to wet
Time and temperature	Workability **falls with time** and falls faster at **higher temperature** (inverse)	Hydration, evaporation and absorption by aggregate. This loss is called **slump loss**
Air entrainment	Entrained air → **more workable**	Tiny air bubbles act like ball bearings and add to paste volume

3 Segregation and bleeding

3.1 Segregation

- **Segregation** is the **separation of the constituents** of a heterogeneous mixture, so that their **distribution is no longer uniform**. In concrete it usually means the coarse aggregate separating from the mortar.
- Concrete must be **cohesive**: **absence of segregation is essential for full compaction**.
- It **depends on the handling and placing** of concrete (e.g. dropping from a height, moving it sideways, long chutes) and on **excessive vibration**.
- It can be **reduced by air entrainment** (which makes the mix more cohesive).
- Result: a **weak layer of laitance** (a layer of fine, watery cement paste) can form on the surface, with honeycombing below.

3.2 Bleeding

- **Bleeding** is a **form of segregation in which water rises to the surface** of freshly placed concrete, because the solids settle and cannot hold all the mixing water.
- Effects:
  - a **weak, non-durable surface** (high w/c at the top, laitance);
  - **plastic shrinkage cracking** when the bleed water evaporates;
  - **poor bond** with reinforcement and between lifts (water collects under bars and coarse aggregate).
- It is **more pronounced in thin slabs** (large surface area for the depth).
- **Reduced by:**
  - **air entrainment**,
  - **rich mixes** (more cement and fines),
  - **adding pozzolans** (fine particles),
  - cement with **high alkali content**, **high C_3A content**, or the addition of **calcium chloride** (all speed up early stiffening).

> **Remember**
>
> Both segregation and bleeding are **controlled by choosing a suitable grading** and by **care in handling and placing** the concrete.

4 Measuring workability

There is **no accepted test that measures workability directly** as defined above. Each test measures something related to workability under particular conditions:

1. **Slump test**
2. **Compacting factor** (and compactability) test
3. **Vebe test**
4. **Flow table test**
5. **Ball penetration (Kelly ball) test**

5 Slump test

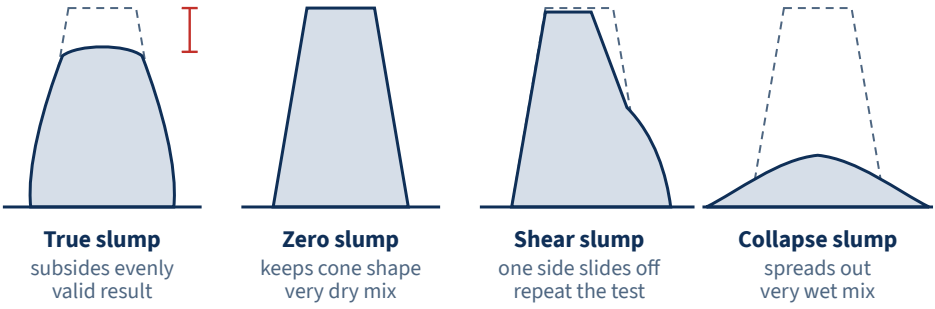
5.1 Apparatus and procedure

- **Mould**: a frustum of a cone, **305 mm high, 203 mm base diameter, 102 mm top diameter** (ASTM C143; IS 1199 uses 300 × 200 × 100 mm), open at both ends, with foot pieces and handles.
- **Tamping rod**: steel, **16 mm diameter**, about 600 mm long, rounded end.
- A non-absorbent base plate and a measuring scale or tape.

Steps:

1. Place the mould on a smooth, level, non-absorbent surface and **stand on the foot pieces**.
2. Fill it in **three layers**, each about one-third of the height, **tamping each layer 25 times** with the rod, evenly over the area.
3. **Strike off** the top level with the rod.
4. **Lift the cone slowly and vertically** (in about 5–10 s), without twisting.
5. The unsupported concrete subsides (**slumps**).
6. Measure the **slump = drop in height** of the concrete, from the top of the mould to the highest point of the slumped concrete.

5.2 Types of slump



Shapes of slump. Only a true slump gives a valid reading.

Type	What happens	Typical slump	Meaning
True slump	Concrete subsides evenly, keeping roughly its shape	Up to about 125 mm	**Valid result**
Zero slump	Concrete keeps the shape of the cone	0	**Very dry** mix (use Vebe or CF test instead)
Shear slump	One side **shears off and slides** down an inclined plane	25–50 mm lateral; up to about 150 mm	Lack of cohesion; **repeat the test**. If it shears again, the mix is harsh
Collapse slump	Concrete **collapses completely** and spreads	150–250 mm	**Very wet** or flowing mix

The measured slump is compared with a table (such as section 6.4) to judge the degree of workability.

5.3 Limitations

- **Not suitable** for concrete with **aggregate larger than 40 mm**.
- **Not suitable for very dry or very wet** mixes (zero or collapse slump).
- **Not always reliable**, because the slump can take any shape (e.g. shear).
- Also: it is a static test, so two mixes with the same slump can behave quite differently when vibrated.

Despite these, it is the **most widely used site test**, because it is simple, quick and cheap, and it is good at **detecting variations between batches** of the same mix.

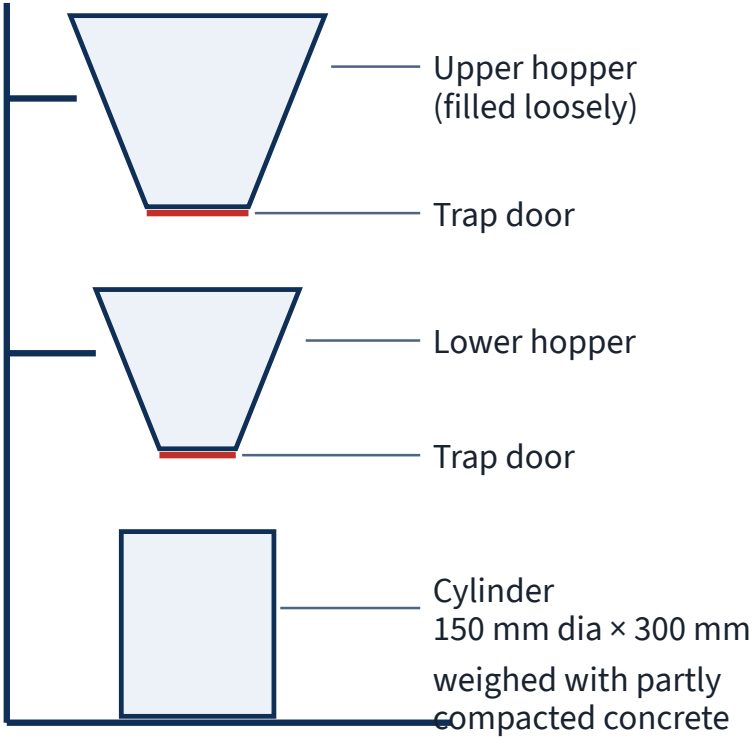
6 Compacting factor test

6.1 Principle

- **Workability can be defined as the amount of work needed to compact the concrete fully** (to its maximum density). Instead of measuring work, the test applies a **standard amount of work** (a fall through hoppers) and measures **how well the concrete is compacted** by it.
- **Compacting factor (CF)** = degree of compaction achieved = **density ratio**: the density actually achieved in the test ÷ the density of the same concrete when

fully compacted.

- Mainly a **laboratory** test; gives **fairly good results for concrete of low workability**.



Compacting factor apparatus. Upper hopper 254 mm top dia. × 279 mm high; lower hopper 229 mm × 229 mm; cylinder about 152 mm dia. × 305 mm (IS 1199).

6.2 Procedure

1. Fill the **upper hopper** gently with concrete (no compaction).
2. **Open its trap door**; the concrete falls into the **lower hopper**.
3. **Open the lower trap door**; the concrete falls into the **cylinder**.
4. Strike off the excess level with the top and **weigh the cylinder with this partially compacted concrete**.

5. Empty the cylinder and **refill it with the same concrete in layers not more than 5 cm thick**, compacting each layer fully (by rodding or vibration) to **100% compaction**.
6. **Weigh the cylinder with the fully compacted concrete**.

Compacting factor

$$\text{CF} = \frac{\text{mass of partially compacted concrete}}{\text{mass of fully compacted concrete}}$$

$$\text{or, equivalently, CF} = \frac{\text{density of partially compacted concrete}}{\text{density of fully compacted concrete}}$$

(Both masses are net of the cylinder.) CF is always less than 1; **a higher CF means higher workability**.

6.3 Merits

- Suitable for **testing workability in the laboratory**.
- Suitable for **concrete of low workability** (where slump is unreliable); it is **sensitive at the low-workability end**.
- Can detect **variations in workability over a wide range**.
- Results are **more precise and sensitive** than the slump test.

6.4 Workability, slump and compacting factor

For concrete with 19 or 38 mm maximum aggregate size (Building Research Establishment):

Degree of workability	Slump (mm)	Compacting factor	Suitable for
Very low	0–25	0.78	Roads vibrated by **power-operated** machines; at the more workable end, sometimes compacted with hand-operated machines

Degree of workability	Slump (mm)	Compacting factor	Suitable for
Low	25–50	0.85	Roads vibrated by **hand-operated** machines; at the more workable end, roads compacted manually with rounded or irregular aggregate; **mass concrete foundations** without vibration or **lightly reinforced** sections with vibration
Medium	50–100	0.92	At the less workable end, manually compacted **flat slabs** with crushed aggregate; **normal reinforced concrete** compacted manually, and **heavily reinforced** sections with vibration
High	100–175	0.95	Sections with **congested reinforcement**; not normally suitable for vibration

7 Vebe test

7.1 Principle and apparatus

- Suitable for **dry concrete of very low workability** (where slump is zero and CF is not sensitive enough).
- Measures the **time needed to remould** a slump-cone sample into a cylinder by **vibration**, i.e. the **effort** the concrete needs to be compacted.
- Apparatus (Vebe consistometer): a **vibrating table**, a **cylindrical container** (about 240 mm dia., 200 mm high) clamped on it, a **slump cone** that fits inside, a **funnel**, and a **transparent disc (rider)** on a vertical rod that slides through a guide on a swivel arm.
- The table vibrates at **50 Hz** with a vertical amplitude of about **±0.35 mm**.

7.2 Procedure

1. Place the slump cone inside the container and **fill it with concrete** through the funnel, as in the slump test.
2. **Remove the slump mould** and swing the stand round so that the **transparent disc rests on top of the concrete**.
3. **Start the vibrator** under the container (and a stopwatch).

4. The vibration **remoulds the concrete**, which spreads to fill the cylindrical container.
5. **Stop** when the **whole underside of the transparent disc is covered** with grout, i.e. the concrete surface has become **horizontal**.
6. The time in seconds is the **Vebe time (Vebe degrees or seconds)**. **Longer time = less workable concrete.**

7.3 Consistency classes

ACI 211.3R recommended values:

Consistency	Slump (mm)	Slump (in.)	Vebe time (s)
Extremely dry	–	–	32 to 18
Very stiff	–	–	18 to 10
Stiff	0 to 25	0 to 1	10 to 5
Stiff plastic	25 to 75	1 to 3	5 to 3
Plastic	75 to 125	3 to 5	3 to 0
Very plastic	125 to 190	5 to 7½	–

IS 1199:1959 classification:

Consistency	Vebe degrees (s)	Characteristics
Moist earth	40 to 25–20	Coarse aggregate particles are adhesive, but the concrete does not clot. **Risk of segregation**
Very dry	20 to 15–10	Like very stiff porridge; forms a stiff mound when dumped and barely shakes or rolls itself into a level surface when carried a long way, say in a wheelbarrow
Dry	10 to 7–5	Like stiff porridge; forms a mound when dumped, and shakes or rolls itself into a level surface when carried a long way in a wheelbarrow
Plastic	5 to 4–3	Can be shaped into a ball between the palms, and sticks to the skin

Consistency	Vebe degrees (s)	Characteristics
Semi-fluid	3 to 2–1	Cannot be rolled into a ball, but spreads out (even if slowly) without losing cohesion, so no segregation
Fluid	More fluid than 1	Spreads out rapidly, and **segregation** occurs

8 Flow table test

8.1 Use

- For concrete of **high workability**, especially **flowing concrete made with superplasticizers**, where the slump test gives a collapse slump.
- Gives a **flow** value: the **arithmetic mean of the diameters** of the spread concrete after the table has been jolted.

8.2 Apparatus

A flat circular **flow table** (brass or steel top) that can be raised and dropped through a fixed height by a **cam and handle**, a **conical mould** (smaller than the slump cone) and a tamping rod.

8.3 Procedure

1. Clean and wet the table; place the **metal cone at the centre** of the table.
2. Fill the mould in **two layers**, **tamping each layer 25 times**; strike off the excess concrete.
3. **Lift the mould slowly and vertically**, leaving the concrete standing unsupported.
4. **Raise the table by 12.5 mm and drop it** (by turning the handle), **15 times in 15 seconds**.
5. Measure the **diameter of the spread** in two perpendicular directions (horizontally and vertically across the table).
6. The **mean of the two diameters**, in mm, is the **flow** of the concrete.

IS 1199 also expresses it as a percentage: **flow (%) = (spread diameter in cm − 25) / 25 × 100**, where 25 cm is the base diameter of the IS mould.

9 Ball penetration (Kelly ball) test

9.1 Use and apparatus

- A **simple, quick field test** for **routine checking** of workability, done directly on concrete in the formwork or a wheelbarrow.
- **Kelly ball**: a **152 mm diameter steel hemisphere weighing 14 kg**, with a handle and a **graduated stem** sliding through a **frame** that rests on the concrete surface.
- Requirements: the concrete must be **at least 200 mm deep**, and the **least lateral dimension of the container at least 460 mm**.

9.2 Procedure

1. Pour the fresh concrete into a container to a **depth of at least 200 mm** and level the top.
2. Set the apparatus on the concrete so that the **frame rests on the surface**, with the ball at least **23 cm from the container edges** (in the middle).
3. **Release the handle** so the ball sinks into the concrete under its own weight.
4. Read the **depth of penetration** on the graduated scale, to the **nearest 6 mm**.
5. **Repeat three times** at different places and take the **average**.

Deeper penetration means more workable concrete. As a rough guide, the slump is about 1.5–2 times the Kelly ball penetration for a given mix.

10 Comparison of tests

10.1 Which test for which concrete

Test	Best for	Measures	Where
Slump	Medium to high workability (about 25–125 mm)	Subsidence under own weight	Site and lab; mostly site

Test	Best for	Measures	Where
Compacting factor	Low to medium workability	Degree of compaction under a standard fall	Mainly lab
Vebe	Very low workability, dry mixes	Time to remould by vibration	Lab
Flow table	Very high workability, flowing concrete	Spread after jolting	Lab and site
Kelly ball	Medium workability	Penetration of a weighted hemisphere	Site, in place

10.2 How the results relate

- **Compacting factor rises as workability rises.**
- **Vebe time rises as workability falls**, because it measures the effort needed to compact the concrete.
- The slide's graph shows **compacting factor falling as Vebe time increases**, plotted against (Vebe time)^{½}, as a band rather than a line.

> **Wording on the slide**
>
> The slide says "Vebe time is a direct function of workability; compaction factor is reciprocal of workability". Physically, it is the other way round: **Vebe time is a measure of the work needed (so it is inversely related to workability)**, while **CF is directly related to workability**. Write it the way your teacher expects, but understand the direction.

- Such relationships are **specific to a particular mix**, so they are **not generally applicable**. They depend on the **shape and texture of the aggregate**, the **presence of entrained air**, and the **mix proportions**.
- The second graph (Dewar) shows Vebe time, compacting factor and slump plotted against each other for **aggregate/cement ratios of 3, 4.5, 6 and 9**: as slump increases, Vebe time drops sharply and CF approaches 1, but the curves differ for each a/c ratio.

- Practical consequence: **the slump test is good for low-to-high workability but insensitive for stiff mixes; CF and Vebe are sensitive for stiff mixes but not for very wet ones.**

11 Density of fresh concrete

- **Density** = **mass per unit volume** of compacted fresh concrete.
- Measured by **compacting concrete in a standard container of known volume and mass** and weighing it (ASTM C138; BS 1881 Part 107; BS EN 12350-6).
- Used to find the **yield** (volume of concrete produced per batch) and the actual **cement content**.

Yield and cement content

$$V = \frac{C + A_f + A_c + W}{\rho}$$

$$\frac{C}{V} = \rho - \frac{A_f + A_c + W}{V}$$

V = volume of concrete per batch (yield); C, A_f, A_c, W = masses of cement, fine aggregate, coarse aggregate and water in the batch; ρ = measured density of fresh concrete. **Cement content** = mass of cement per unit volume of concrete = C/V.

Worked example

A batch contains 50 kg cement, 100 kg fine aggregate, 180 kg coarse aggregate and 25 kg water. The measured density of the fresh concrete is 2400 kg/m^3. Find the yield and the cement content.

- Total mass = 50 + 100 + 180 + 25 = 355 kg
- Yield V = 355 / 2400 = **0.148 m^3** per batch
- Cement content = C/V = 50 / 0.1479 = **338 kg/m^3**
- Check with the second formula: 2400 − (100 + 180 + 25)/0.1479 = 2400 − 2062 = 338 kg/m^3 ✓

The measured density also tells you the **air content** if the theoretical air-free density is known: air (%) = (theoretical density − measured density) / theoretical density × 100.

Quick revision

- Workability: ease of mixing, placing, compacting and finishing; also called consistency. 5% voids → about 30% loss of strength.
- Factors: water content (most important), aggregate shape/texture/grading, a/c ratio (inverse), admixtures, cement fineness, time and temperature (slump loss), air entrainment.
- Segregation: separation of constituents; from bad handling, over-vibration; reduced by air entrainment; causes laitance. Bleeding: water rises; weak surface, plastic shrinkage cracks, poor bond; worse in thin slabs; reduced by air entrainment, rich mixes, pozzolans, high alkali / high C_3A cement, $CaCl_2$.
- Slump cone 305 × 203 × 102 mm (IS 300 × 200 × 100); 3 layers × 25 strokes, 16 mm rod. True, zero, shear (repeat), collapse. Not for aggregate > 40 mm or very dry/wet mixes.
- CF = partially compacted mass / fully compacted mass; lab, low workability, more sensitive. Very low 0.78, low 0.85, medium 0.92, high 0.95.

- Vebe: very dry mixes; 50 Hz, ±0.35 mm; time for disc to be fully covered = Vebe seconds; longer = stiffer.
- Flow table: flowing concrete; 2 layers × 25; 15 drops of 12.5 mm in 15 s; mean spread diameter.
- Kelly ball: 152 mm, 14 kg; depth ≥ 200 mm, width ≥ 460 mm, 23 cm from edges; read to 6 mm; average of 3.
- Yield $V = (C + Af + Ac + W)/\rho$; cement content = C/V.

Questions to practise

1. Define workability. Why is it important? Explain the relation between density ratio and strength ratio. (5 marks)
2. Discuss the factors affecting the workability of concrete. (6 marks)
3. What are segregation and bleeding? State their causes, effects and how they can be controlled. (8 marks)
4. Describe the slump test with a sketch of the different types of slump. What are its limitations? (8 marks)
5. Describe the compacting factor test. Why is it preferred to the slump test for low-workability concrete? (8 marks)
6. Describe the Vebe consistometer test. For what type of concrete is it used? (6 marks)
7. Write short notes on the flow table test and the Kelly ball test. (6 marks)
8. Compare the slump, compacting factor and Vebe tests. (5 marks)
9. In a compacting factor test, the mass of partially compacted concrete was 11.8 kg and that of fully compacted concrete was 12.6 kg. Find the compacting factor and state the degree of workability. (3 marks)
10. A batch uses 50 kg cement, 95 kg sand, 190 kg coarse aggregate and 27.5 kg water, and the fresh concrete density is 2380 kg/m^3. Find the yield per batch and the cement content. (4 marks)